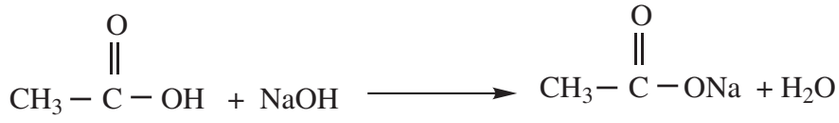
Example

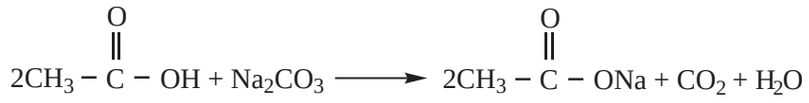


Accteic acid — Sodium acetate

3) Reaction with carbonates and bicarbonates (Test for carboxylic acid group)

Carboxylic acids decompose carbonates and bicarbonates evolving carbondioxide gas with effervescence.

Example



Acetic acid — Sodium acetate

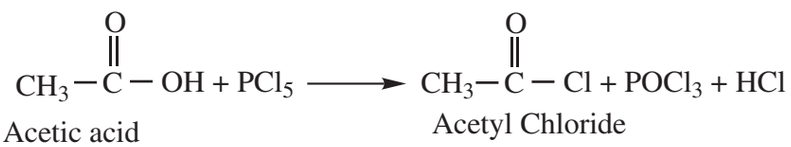
4) All Carboxylic acids turn blue litmus red

B) Reactions involving cleavage of C-OH bond

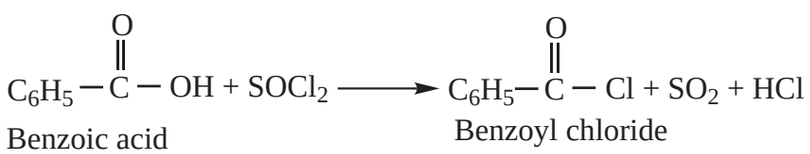
1) Reactions with PCl_5, PCl_3 and $SOCl_2$

The hydroxyl group of carboxylic acids behaves like that of an alcoholic group and is easily replaced by chlorine atom on treating with PCl_5, PCl_3 or $SOCl_2$.

Example



Acetic acid — Acetyl Chloride

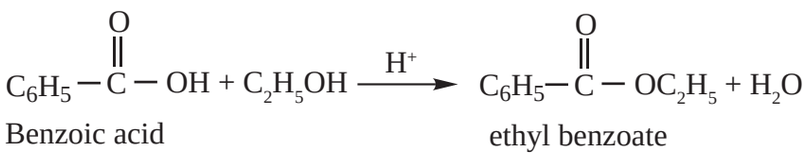


Benzoic acid — Benzoyl chloride

2) Reactions with alcohols (Esterification)

When carboxylic acids are heated with alcohols in the presence of conc. H_2SO_4 or dry HCl gas, esters are formed. The reaction is reversible and is called esterification.

Example



Benzoic acid — ethyl benzoate

XII U12-Carbonyl Compounds.indd 173 11-01-2021 16:10:42